

8.5 Integration of Rational Functions by Partial Fractions

This section shows how to express a rational function (a quotient of polynomials) as a sum of simpler fractions, called *partial fractions*, which are easily integrated. For

instance, the rational function $\frac{5x-3}{x^2-2x-3}$ can be rewritten as

$$\frac{5x-3}{x^2-2x-3} = \frac{2}{x+1} + \frac{3}{x-3}.$$

To integrate the rational function $\frac{5x-3}{x^2-2x-3}$ on the left side of our previous expression,

we simply sum the integrals of the fractions on the right side:

$$\begin{aligned}\int \frac{5x-3}{x^2-2x-3} dx &= \int \frac{2}{x+1} dx + \int \frac{3}{x-3} dx \\ &= 2\ln|x+1| + 3\ln|x-3| + C.\end{aligned}$$

◆ Method of Partial Fractions When $f(x)/g(x)$ Is Proper

Success in writing a rational function $f(x)/g(x)$ as a sum of partial fractions depends on two things:

- The degree of $f(x)$ must be less than the degree of $g(x)$.
- We must know the factors of $g(x)$.

Here is how we find the partial fractions of a proper fraction $f(x)/g(x)$ when the factors of g are known.

(a) If $g(x)$ has a factor $(x-r)^m$, then

$$\frac{f(x)}{g(x)} = \frac{A_1}{x-r} + \frac{A_2}{(x-r)^2} + \cdots + \frac{A_m}{(x-r)^m} + \cdots.$$

(b) If $g(x)$ has a factor $(x^2+px+q)^n$, then

$$\frac{f(x)}{g(x)} = \frac{B_1x+C_1}{x^2+px+q} + \frac{B_2x+C_2}{(x^2+px+q)^2} + \cdots + \frac{B_nx+C_n}{(x^2+px+q)^n} + \cdots.$$

For example,

$$\frac{x^2+3x+1}{(x+1)(x+2)(x+3)} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{x+3}.$$

$$\frac{x^2+3x+1}{(x+1)^2(x^2+2x+7)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{Cx+D}{x^2+2x+7}.$$

Example 1 Use partial fractions to evaluate

$$\int \frac{x^2+4x+1}{(x-1)(x+1)(x+3)} \, dx.$$

Example 2 Evaluate $\int \frac{6x+7}{(x+2)^2} \, dx$.

Example 3 Evaluate $\int \frac{-2x+4}{(x^2+1)(x-1)^2} \, dx$.

Example 4 Evaluate $\int \frac{1}{x(x^2+1)^2} \, dx$.

Example 5 Evaluate $\int \frac{1}{x^3+x^2-2x} \, dx$.